

TASK6: #51630

1. Research and learn about the error's rises regarding with website hosting.
2. study well about the error's within the error code range 300 - 500.

eg: 500- internal server error

404- file not found error

3.also note down the cause of this errors,as well as the trouble shooting methods.

1. Introduction

When a browser requests a webpage, the web server returns an HTTP status code that describes the result of the request. Codes in the 300–399 range generally indicate redirection, 400–499 indicate that the request could not be fulfilled because of a client/request-related condition, and 500–599 indicate a server-side failure or inability to complete a valid request.

2. HTTP Status Code Ranges

Range	Class	General Meaning
300–399	3xx Redirection	The client must take additional action, commonly following a redirect.
400–499	4xx Client / Request Error	The request is invalid, unauthorized, forbidden, missing, or otherwise cannot be served.
500–599	5xx Server Error	The server or an upstream service failed while handling an otherwise valid request.

3. Common 3xx Errors and Redirect Responses

Code	Meaning	Common Causes in Hosting	Troubleshooting

300	Multiple Choices	Server has more than one possible representation for the requested resource; uncommon in normal hosting setups.	Inspect the response headers/body and server configuration. Check content negotiation rules and application routing.
301	Moved Permanently	HTTP-to-HTTPS redirect, domain migration, changed URL, CMS redirect, or web-server rewrite rule.	Check Location header and redirect chain. Review Apache .htaccess, Nginx config, CMS redirects, and canonical URL settings.
302	Found / Temporary Redirect	Temporary redirect rules, application redirects, login flows, or misconfigured rewrite rules.	Trace the full redirect chain. Confirm whether the redirect is intentional and inspect application/server logs.
303	See Other	Application uses a redirect after a POST or other request to point the client to another resource.	Check application routing and Location header. Verify the destination URL and expected request method.
304	Not Modified	Normal browser/proxy caching behavior when the resource has not changed.	Usually not an error. If stale content is suspected, inspect Cache-Control/ETag/Last-Modified headers and CDN cache.
307	Temporary Redirect	Temporary redirect while preserving the original HTTP method.	Check redirect rules and destination. Make sure POST/PUT requests are intentionally preserved.
308	Permanent Redirect	Permanent redirect while preserving the original HTTP method.	Check web-server/CMS redirect configuration and verify that the destination and method behavior are correct.

Important: 3xx responses are not necessarily errors. A redirect becomes a troubleshooting problem when it is unexpected, forms a loop, points to the wrong URL, or creates an excessive redirect chain.

4. Common 4xx Errors

Code	Meaning	Common Causes	Troubleshooting
400	Bad Request	Malformed URL/request, invalid headers, oversized or invalid request data, application request parsing failure.	Check the exact request URL, headers, cookies and payload. Review reverse-proxy/web-server and application logs. Test with curl or browser developer tools.
401	Unauthorized	Authentication is required or credentials/token are missing, invalid, or expired.	Verify login/session/token configuration, Authorization headers, cookies, and authentication middleware. Check server logs.
402	Payment Required	Reserved for future use; some services use it for subscription/payment restrictions.	If a hosting/API provider uses it, check the provider's billing or service documentation and account status.
403	Forbidden	Permissions, security rules, IP restrictions, WAF rules, directory access restrictions, or authentication/authorization policy.	Check file/directory permissions and ownership, web-server access rules, .htaccess/Nginx rules, WAF/CDN events, and application authorization.
404	Not Found	Requested file/page does not exist, wrong URL, broken link, missing deployment file, or incorrect routing.	Confirm the URL and case. Check document root/deployment output, rewrite rules,

			CMS routes, symlinks and application logs.
405	Method Not Allowed	The resource exists but does not accept the HTTP method used, such as POST instead of GET.	Inspect the Allow header and route configuration. Verify frontend/API method and web-server rules.
406	Not Acceptable	Server cannot provide a representation matching the request's Accept headers; sometimes caused by security rules.	Inspect Accept/Content negotiation headers and application/server configuration. Test with a simpler request.
408	Request Timeout	Client took too long to send the request or connection timed out.	Check network stability, proxy/load-balancer timeouts, slow uploads, server connection settings, and logs.
409	Conflict	Request conflicts with current resource state, common with APIs or concurrent updates.	Inspect API/application logs and request state. Check duplicate resources, version conflicts and concurrent writes.
410	Gone	Resource was intentionally removed and is not expected to return.	Confirm removal is intentional. Update links or redirects where appropriate.
413	Content Too Large	Request body/upload exceeds a configured limit.	Check web-server, reverse-proxy, CDN and application upload limits. Increase limits only when appropriate.
414	URI Too Long	URL is longer than the server/proxy accepts.	Reduce query-string size; use POST/body data where appropriate and

			inspect proxy/server limits.
415	Unsupported Media Type	Request Content-Type is unsupported by the application.	Send the expected Content-Type and payload format; verify API documentation and server parsing configuration.
416	Range Not Satisfiable	Requested byte range cannot be served, often with downloads/media.	Check Range headers, file size and proxy/cache behavior. Test without a Range header.
429	Too Many Requests	Rate limiting triggered by excessive requests from a client/IP/token.	Review rate-limit configuration and logs. Reduce request frequency, use backoff, and check for bots or loops.

5. Common 5xx Errors

Code	Meaning	Common Causes	Troubleshooting
500	Internal Server Error	Unhandled application exception, PHP/runtime error, bad server configuration, broken rewrite rule, faulty plugin/module, or permission/configuration issue.	Check web-server and application error logs first. Review recent code/config changes, PHP/runtime versions, .htaccess/Nginx syntax, dependencies, permissions and environment variables. Reproduce with a minimal request.
501	Not Implemented	Server does not support the functionality/method requested.	Check whether the server/proxy supports the method or feature. Verify configuration and provider limitations.
502	Bad Gateway	Reverse proxy/load balancer received an	Check upstream application/PHP-

		invalid response from an upstream server.	FPM/container/service health, ports, sockets, DNS, TLS and timeout settings. Inspect proxy and upstream logs.
503	Service Unavailable	Server temporarily unable to handle the request due to overload, maintenance, failed service, resource exhaustion or health-check failure.	Check CPU/RAM/disk, process/service status, deployment/maintenance state, connection pools, health checks and application logs. Restart failed services only after identifying the cause.
504	Gateway Timeout	Proxy/gateway did not receive a timely response from an upstream service.	Identify the slow upstream request. Check application/database latency, network connectivity, proxy and upstream timeouts, queue saturation and slow queries.
505	HTTP Version Not Supported	Server does not support the HTTP protocol version used in the request.	Check client, proxy/CDN and server HTTP-version configuration. Update incompatible components where appropriate.
506	Variant Also Negotiates	Configuration error involving transparent content negotiation.	Review content-negotiation and variant configuration; check for circular negotiation definitions.
507	Insufficient Storage	Server cannot store the representation needed to complete the request.	Check filesystem/object-storage quota, disk space, inode availability and application storage limits.
508	Loop Detected	Server detected an infinite loop while processing a request, often related to WebDAV	Inspect rewrite rules, aliases, mounted paths and application recursion. Trace the

		or recursive configuration.	request path and server logs.
509	Bandwidth Limit Exceeded	Non-standard code sometimes used by hosting providers when an account/site exceeds bandwidth limits.	Check hosting usage and provider-specific limits. Review traffic spikes, large files, hotlinking and CDN configuration.
510	Not Extended	Request requires extensions that the server cannot satisfy.	Inspect protocol extensions and server configuration. This status is uncommon on ordinary websites.
511	Network Authentication Required	Client needs to authenticate to gain network access, often from captive portals.	Check network/captive-portal authentication. This is generally not a website application error.

6. Practical Troubleshooting

1. Identify the exact status code:

Record the URL, time, HTTP method, response headers, browser message, and whether the issue affects one user or many.

2. Check browser developer tools:

Use the Network tab to inspect redirects, request/response headers, status code, timing, cookies and response body.

3. Reproduce outside the browser:

Use a tool such as curl to determine whether the problem is browser-specific or occurs at the HTTP/server level.

4. Check DNS and CDN/proxy:

Confirm the hostname resolves to the intended service and determine whether a CDN, WAF, reverse proxy or load balancer sits in front of the origin.

5. Check web-server logs:

For Apache/Nginx/IIS or a hosting panel, inspect access and error logs at the exact time of failure.

6. Check application/runtime logs:

For PHP, Node.js, Python, Java, .NET or a CMS, inspect application exceptions, dependency failures and runtime errors.

7. Check resources:

For 5xx errors, inspect CPU, RAM, disk space, inodes, process limits, worker pools, database connections and network capacity.

8. Check recent changes:

Compare the timing with deployments, plugin/theme changes, configuration edits, DNS changes, SSL changes and software upgrades.

9. Fix and retest:

Make one controlled change at a time, clear relevant caches, retest the original URL, and verify related URLs or API endpoints.

10. Prevent recurrence:

Add monitoring, log retention, backups, health checks, sensible timeouts/rate limits, and deployment rollback procedures.

7. Causes to Check

- Incorrect document root or virtual-host/server-block configuration.
- DNS record points to the wrong server or old hosting provider.
- Expired domain, suspended hosting account, or provider resource limit.
- SSL/TLS certificate or HTTPS redirect configuration problems.
- Apache .htaccess or Nginx rewrite rules causing redirects, 403s, 404s, or 500s.
- Incorrect Linux file ownership/permissions or missing deployment files.
- PHP-FPM, Node.js, Python, database, container, or other upstream service is stopped or unhealthy.
- Insufficient CPU, memory, disk space, inodes, worker processes, or database connections.
- CDN/WAF/security rules blocking legitimate requests or caching an unwanted response.
- Application bugs, missing environment variables, incompatible dependencies, or failed migrations.

8. Commands To Check

curl -I https://example.com/ — View response headers and status code.

curl -v https://example.com/path — Show detailed request/connection information.

curl -L -I https://example.com/ — Follow redirects and inspect the redirect chain.

dig example.com — Inspect DNS resolution.

nslookup example.com — Check DNS resolution on many systems.

9. SUMMARY

Code	Meaning	First Things to Check
301/302	Redirect	Redirect rules, Location header, CMS, HTTPS/domain settings
304	Not Modified	Cache headers; usually normal
400	Bad Request	URL, headers, cookies, request body
401	Unauthorized	Credentials, token, session/authentication
403	Forbidden	Permissions, WAF, access rules
404	Not Found	URL, document root, deployment, routes/rewrite rules
405	Method Not Allowed	HTTP method and route configuration
408	Request Timeout	Network and timeout settings
413	Content Too Large	Upload/request size limits
429	Too Many Requests	Rate limits and traffic patterns
500	Internal Server Error	Application/server logs, configuration, runtime
502	Bad Gateway	Upstream service, proxy, PHP-FPM/container, ports
503	Service Unavailable	Service health, resource limits, maintenance
504	Gateway Timeout	Slow upstream, database, timeout/queue settings
507	Insufficient Storage	Disk/quota/inodes/storage